
BY 505: Nanobiotechnology

Course Code	BY 505
Course Name	Nanobiotechnology
L-T-P-C	3-0-0-3
Prerequisites	IC136, or consent of faculty
Intended for	3rd and 4th year UG and PG
Distribution	Elective
Approval	8th Senate (Oct 2015)

Course Contents

- **Introduction**

Definition of nanotechnology, small-strange and useful, why go nano – unique properties of nanomaterials, history of nanotechnology, present and future of nanotechnology, Nano + Light, Engineering optical properties, Band gaps, exciton, quantum confinement, Different kinds of nanomaterials- Metal and semiconductor nanoparticles; Polymeric nanoparticles; Molecular nanoparticles, Forces at the nanoscale, The Nano-Bio interface. [15 Lectures]

- **Nanobiosensing** Definition of sensors, different elements of sensor, introduction to nanobiosensing, Different types of biosensors, surface plasmon resonance based biosensor, electrochemical and potentiometric based biosensor, motion, temperature, chemical, light and pressure sensitive biosensors, Applications of biosensors in molecule analysis; food safety, environmental and biomedical monitoring and detection of biological weapons, Lab on chip devices for sensing and detection. [15 Lectures]

- **Nanomedicine**

Nanoparticle within a biological environment, Nanoparticle dynamics in biological media, nanoparticles for therapy- drug delivery, gene delivery, protein delivery, photothermal and photodynamic therapy, uptake and toxicology of nanomaterials, Nanomaterials for tissue engineering and prosthetics. Nanoparticles based bioimaging. [12 Lectures]

Text and Reference Books

1. Jeremy Ramsden, **Nanotechnology: An Introduction**, Elsevier Publishers, 2011.
2. C.M. Niemeyer and C. A. Mirkin (Editors), **Nanobiotechnology: Concepts, Applications and Perspectives**, Wiley-VCH Verlag GmbH & Co, 2012.
3. Huw Summers (Editor), **Nanomedicine**, Elsevier Publishers, 2013.
4. Sandro Carrara (Editor), **Nano-Bio-Sensing**, Springer Publishers, 2011.
5. Additionally, other latest research articles related to the topic will be discussed.